

BIRNI NKONNI, Niger

WMO: 610750

Lat: 13.800N

Lon: 5.250E

Elev: 273

StdP: 98.09

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.9	17.0	-7.5	2.1	27.1	-5.8	2.4	27.1	8.1	24.6	7.1	24.3	1.1	100	0.375

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	14.2	42.6	21.5	41.7	21.6	40.8	21.5	26.6	34.7	26.2	34.2	26.0	33.8	2.0	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.8	20.5	29.4	24.4	20.0	29.2	24.1	19.6	29.1	84.9	34.6	83.3	34.3	81.9	34.0	30.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.1	5.0	4.2	DB	13.2	44.5	1.0	0.5	12.5	44.8	11.9	45.1	11.3	45.4	10.5	45.8	(4)
(5)			WB	5.8	27.5	0.8	0.9	5.3	28.1	4.8	28.7	4.4	29.2	3.8	29.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	29.8	24.8	28.1	31.7	34.4	34.5	32.2	29.7	28.2	29.4	30.9	28.6	25.4	(6)
(7)		DBStd	3.56	2.29	2.66	2.15	1.55	1.88	2.20	1.97	1.62	1.82	1.33	1.74	2.07	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	7232	457	506	672	731	761	664	609	564	583	647	559	479	(10)
(11)		CDD18.3	4190	199	272	413	481	502	414	351	306	333	389	309	221	(11)
(12)		CDH23.3	57786	2061	3466	6282	7911	8156	6199	4569	3467	4129	5303	3929	2314	(12)
(13)		CDH26.7	34846	982	2001	4116	5563	5702	3898	2371	1499	2104	3141	2335	1134	(13)
(14)	Wind	WSAvg	1.5	1.8	1.8	1.7	1.5	1.8	2.0	1.8	1.3	1.1	1.0	1.1	1.7	(14)
(15)	Precipitation	PrecAvg	482	0	1	1	4	28	64	129	164	81	12	0	0	(15)
(16)		PrecMax	643	5	50	12	38	113	157	285	345	319	106	0	1	(16)
(17)		PrecMin	290	0	0	0	0	0	8	38	0	16	0	0	0	(17)
(18)		PrecStd	97	1	7	2	8	26	35	53	63	50	21	0	0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.4	40.9	42.4	43.8	43.4	40.9	38.0	35.7	38.1	39.9	39.2	37.1	(19)
(20)			MCWB	17.7	19.0	19.6	21.0	23.0	24.7	25.2	25.9	25.5	21.1	19.6	18.5	(20)
(21)		2%	DB	35.3	39.1	41.5	42.9	42.3	39.6	36.6	34.4	36.6	38.9	38.4	35.5	(21)
(22)			MCWB	16.9	18.6	19.2	20.9	23.4	24.7	25.0	25.6	25.6	21.6	19.3	17.5	(22)
(23)		5%	DB	33.7	37.6	40.6	42.1	41.3	38.5	35.5	33.5	35.5	38.0	37.4	34.1	(23)
(24)			MCWB	16.1	18.1	19.0	21.2	23.6	24.8	25.0	25.3	25.6	22.1	18.9	16.7	(24)
(25)		10%	DB	32.1	35.8	39.4	41.1	40.3	37.4	34.4	32.5	34.3	37.0	36.2	32.5	(25)
(26)			MCWB	15.5	17.5	18.7	21.1	23.8	24.7	24.8	25.1	25.5	22.5	18.3	16.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.2	22.0	23.2	25.6	26.8	27.1	26.6	26.6	27.2	26.6	22.4	19.1	(27)
(28)			MCDB	34.7	34.7	36.9	38.8	38.3	36.5	34.0	33.4	34.1	34.9	35.3	35.4	(28)
(29)		2%	WB	17.7	20.2	21.9	24.8	26.1	26.2	26.0	26.2	26.6	26.1	20.6	18.3	(29)
(30)			MCDB	33.8	35.7	36.6	37.7	37.4	35.6	33.8	32.5	33.6	34.0	34.9	34.1	(30)
(31)		5%	WB	16.6	18.9	20.2	24.3	25.6	25.8	25.6	25.8	26.2	25.6	19.7	17.4	(31)
(32)			MCDB	32.5	35.8	38.2	36.9	36.9	35.0	33.0	31.7	33.1	33.4	34.8	32.8	(32)
(33)		10%	WB	15.7	17.9	19.2	23.7	25.2	25.4	25.3	25.5	25.9	25.1	19.0	16.4	(33)
(34)			MCDB	31.3	35.1	38.1	36.2	36.4	34.4	32.3	31.0	32.6	32.7	34.5	31.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	14.7	15.0	15.4	14.2	11.8	10.3	8.9	8.2	9.9	13.2	15.6	14.5	(35)
(36)			MCDBR	16.8	16.9	16.9	15.4	13.2	11.2	10.2	9.6	11.3	14.7	16.7	16.4	(36)
(37)			MCWBR	7.1	6.6	6.4	5.9	4.2	2.8	2.6	2.8	3.3	4.2	5.7	6.7	(37)
(38)		5% WB	MCDBR	16.5	16.1	16.0	12.7	11.3	10.5	9.4	9.0	10.5	11.7	16.6	16.3	(38)
(39)			MCWBR	7.3	6.9	7.0	5.0	3.6	3.2	3.0	3.1	3.6	3.5	6.0	6.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.579	0.636	0.704	0.735	0.725	0.699	0.634	0.603	0.593	0.621	0.544	0.539	(40)	
(41)		taud	1.720	1.570	1.456	1.469	1.509	1.582	1.760	1.817	1.823	1.700	1.759	1.779	(41)	
(42)		Ebn at Noon	722	702	669	649	644	653	700	731	740	705	745	741	(42)	
(43)		Edh at Noon	229	276	316	312	294	270	227	217	216	239	218	211	(43)	
(44)	All-Sky Solar Radiation	RadAvg	5.48	6.00	6.32	6.50	6.33	6.05	5.90	5.65	6.15	6.07	5.74	5.37	(44)	
(45)		RadStd	0.24	0.25	0.34	0.24	0.20	0.25	0.19	0.23	0.20	0.17	0.14	0.21	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page